

Resistor and resistance determination

When you run the program, a window similar to that in Fig. 1 appears. There are two tabs in the main menu. The *Resistors* tab is shown by default each time you open the program.

The first option allows the user to select the number of significant figures required (two or three).

If you select 2, the resistor will have four colour bands (or five, if temperature coefficient is set). The first two colour bands represent the resistance value corrected to two significant figures, the third is the decimal multiplier, the fourth band represents the tolerance percentage and the optional fifth band shows temperature coefficient.

When 3 is selected, the resistor can have five (or six) colour bands. The entered resistor value is corrected to three significant figures and third band representing the third significant figure is added to the resistor, followed by the fourth band representing the decimal. The next few options allow you to enter the data – resistance value, tolerance and temperature coefficient of the resistor to be used. You can also find the resistor value based on the colour code by selecting the colour of each band in the resistor from the drop-down menu provided.

Refer the colour code system shown in Fig. 2 for details on the position of bands and all the possible colour coding.

Resistor Colour-Coding									
4-Band Color Code				250Ω 25%					
5-Band Color Code				460Ω 21%					
6-Band Color Code				270Ω 20%					
1st Digit	2nd Digit	3rd Digit	Multiplier	Tolerance					
0	0	0	0.01 Silver	±10% Gold					
1	1	1	0.1 Gold	±5% Green					
2	2	2	1	±2% Blue					
3	3	3	10	±1% Brown					
4	4	4	100	±0.5% Red					
5	5	5	1000	±0.25% Orange					
6	6	6	10000	±0.1% Yellow					
7	7	7	1M	±0.05% Purple					
8	8	8	10M	±0.01% Grey					
9	9	9							

Fig 2: Colour code system of resistors

The next part of the window lists the names and corresponding resistance values of all the standard resistors in E series and Renard's numbers that are closest to and compatible with the value of resistor you selected. By clicking on the *Take over* button near each resistor, the entered values and their colour is replaced by the values of the selected standard resistor.

Example. Suppose the resistor value required is 178kΩ, with two significant digits and 10 per cent tolerance. First type 178 in the box for resistor value. This automatically corrects to 170. Now append three zeros to it. The bands of the resistor depicted below will change instantly to the corresponding colour colours. Just below this, each band and their colours will be displayed. First band – brown, second band – violet, fourth band – yellow, fifth band – grey.

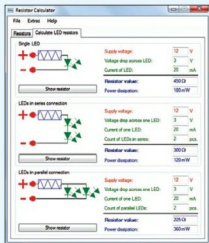


Fig 3: Calculating resistance values for LEDs

Third band will be empty because there are no three significant figures. Sixth band is also left blank since we have not selected a temperature coefficient.

Below this section, all the resistors in E series and Renard's numbers compatible to the entered value are listed. The closest match given in this set would be highlighted with green font. In the above example it is the E series resistor E12: 180Ω.

Now, if you click on the arrow near each colour given for each band, it opens a drop-down menu which lists all the colours as in the standard code.

Suppose we select first band – orange, second band – white, fourth band – red, fifth band – blue and sixth band – black. The resistor value changes to 3900, tolerance to ±0.25 per cent and temperature coefficient to 200 ppm

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